Lecture 22: Summary of Curve Sketching

Goal: Use the tools of calculus to draw a more accurate graph of a function.
All examples come from §4.5 of James Stewart's *Calculus Early Transcendentals*, 5th Edition as well as §4.4 of *Calculus Early Transcendentals*, 3rd Edition by Briggs et. al. (I'm too lazy to make my own examples).

All graphs will be available on Desmos (here).

Guidelines for Sketching a Curve

For drawing the graph of $y = f(x)$ by hand, consider:

(A) **Domain:** Find all values where f is defined.

(B) **Intercepts:** Find the y-intercept by computing $f(0)$. Find the x-intercepts by solving the equation $f(x) = 0$.

(C) **Symmetry:**



(i) *Even functions:* When $f(x) = f(-x)$; this means the graph is symmetric about the y-axis.

(ii) *Odd functions:* When $f(-x) = -f(x)$; this means it is symmetric about the origin (the left side is a 180° rotation of the right side)

(iii) *Periodic functions:* When $f(x+p) = f(x)$ for some $p > 0$, the smallest such p is called the period. You only need to focus on one period of the function.

(D) **Asymptotes and End Behavior:**



(i) *Vertical Asymptotes:* Occurs at $x = a$ when either:

$$\lim_{x \to a^-} f(x) \text{ OR } \lim_{x \to a^+} f(x)$$

is not finite (equals ∞ or $-\infty$).

(ii) *Horizontal Asymptotes:* Occurs at $y = L$ when either:

$$\lim_{x \to -\infty} f(x) = L \quad \text{OR} \quad \lim_{x \to \infty} f(x) = L$$



(iii) *Slant Asymptotes:* The line $y = mx + b$ is a slant asymptote when either :

$$\lim_{x \to -\infty} [f(x) - (mx + b)] = 0 \quad \text{OR} \quad \lim_{x \to \infty} [f(x) - (mx + b)] = 0$$

For rational functions, if the quotient from long division $q(x)$ is linear, then $q(x)$ is the slant asymptote.

(E) **Intervals of Increase or Decrease:** Use the I/D Test to see when f is increasing or decreasing. If $f' > 0$, then f is increasing. If $f' < 0$, f is decreasing.

(F) **Relative Maximum and Minimum Values:** Use the 1st or 2nd derivative test to determine if critical numbers are relative extrema.

(G) **Concavity and Inflection Points:** Use the concavity test. If $f'' > 0$, then f is concave upwards. If $f'' < 0$, f is concave downwards.

(H) **If needed, get more information:** You can also compute $(x, f(x))$ pairs to see the height of the graph. The tangent line also tells you where the graph is going at that point.





Example 1 (Why Graphing Calculators Can Be Deceiving)**.** Sketch a graph of $f(x) = \frac{1}{3}x^3 - 400x$

Domain: $(-\infty, \infty) = \mathbb{R}$

y-int: $f(0) = \frac{1}{3}\cdot 0^3 - 400\cdot 0 = 0$

x-int: $\frac{1}{3}x^3 - 400x \overset{\text{set}}{=} 0$

$x^3 - 1200x = 0$

$x(x^2 - 1200) = 0$

$\Rightarrow x = 0, \pm\sqrt{1200}$

$\Rightarrow x \approx 0, \pm 34.6$

Asymptotes and End Behavior:

$\lim_{x\to\infty}\left[\frac{1}{3}x^3 - 400x\right] = \lim_{x\to\infty}\frac{1}{3}x^3 = \infty$

$\lim_{x\to-\infty}\left[\frac{1}{3}x^3 - 400x\right] = \lim_{x\to-\infty}\frac{1}{3}x^3 = -\infty$

ID and C Test:

$f(x) = \frac{1}{3}x^3 - 400x$

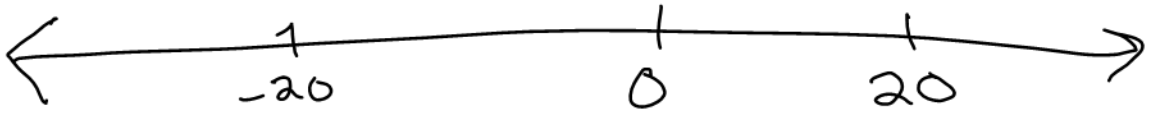
$f'(x) = x^2 - 400 \overset{\text{set}}{=} 0$

$x^2 = 400$

$x = \pm 20$

$f''(x) = 2x \overset{\text{set}}{=} 0$

$x = 0$

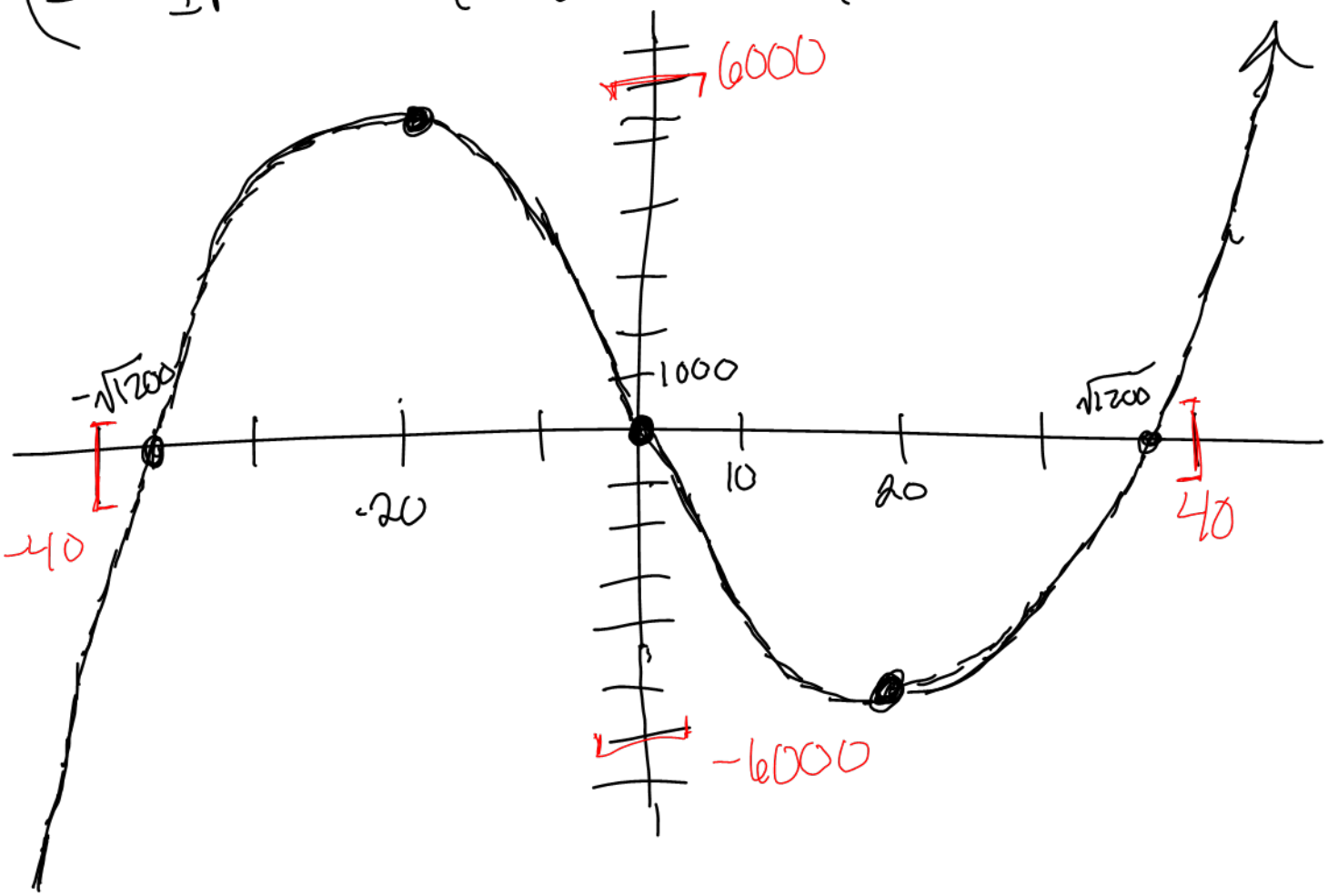


Test Points	-100	-1	1	100
Sign of f'	$+$	$-$	$-$	$+$
Sign of f''	$-$	$-$	$+$	$+$
I/D Test	Inc ↗	↘ Dec	Dec ↘	↗ Inc
C Test	CD	CD	CU	CU

(1) Rel. max at $(-20, f(-20)) = (-20, 5333\frac{1}{3})$

(2) Rel. min at $(20, f(20)) = (20, -5333\frac{1}{3})$

(3) IP at $(0, f(0)) = (0,0)$



Example 2. Sketch a graph of $f(x) = \dfrac{2x^2}{x^2-1}$

$\dfrac{2x^2}{x^2-1}$

VAs and Domain:

x^2-1 set 0

$x = \pm 1$

Domain: $(-\infty,-1) \cup (-1,1) \cup (1,\infty) = \mathbb{R}-\{-1,1\}$

HAs and End Behavior:

$\lim\limits_{x\to\infty} \dfrac{2x^2}{x^2-1} = \lim\limits_{x\to\infty} \dfrac{2x^2}{x^2} = 2$

$\Rightarrow$ $y=2$ is a HA of f

Intercepts: Only passes through the origin

I/D and C Test:

$f(x) = \dfrac{2x^2}{x^2-1}$

$f'(x) = \dfrac{4x(x^2-1) - 2x^2(2x)}{(x^2-1)^2} = \dfrac{4x^3-4x-4x^3}{(x^2-1)^2}$

$= \dfrac{-4x}{(x^2-1)^2}$ set 0

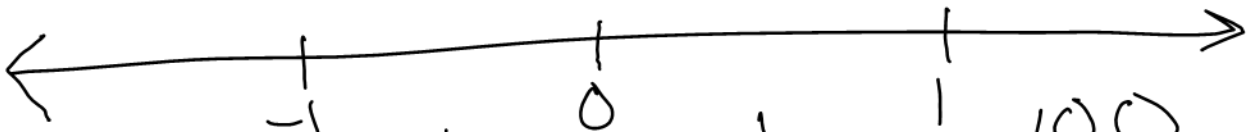
$-4x = 0$

$x = 0$

$$f''(x) = \frac{-4(x^2-1)^2 - (-4x)(2)(x^2-1)(2x)}{(x^2-1)^4}$$

$$= \frac{-4(x^2-1) + 16x^2}{(x^2-1)^3} = \frac{12x^2+4}{(x^2-1)^3}$$

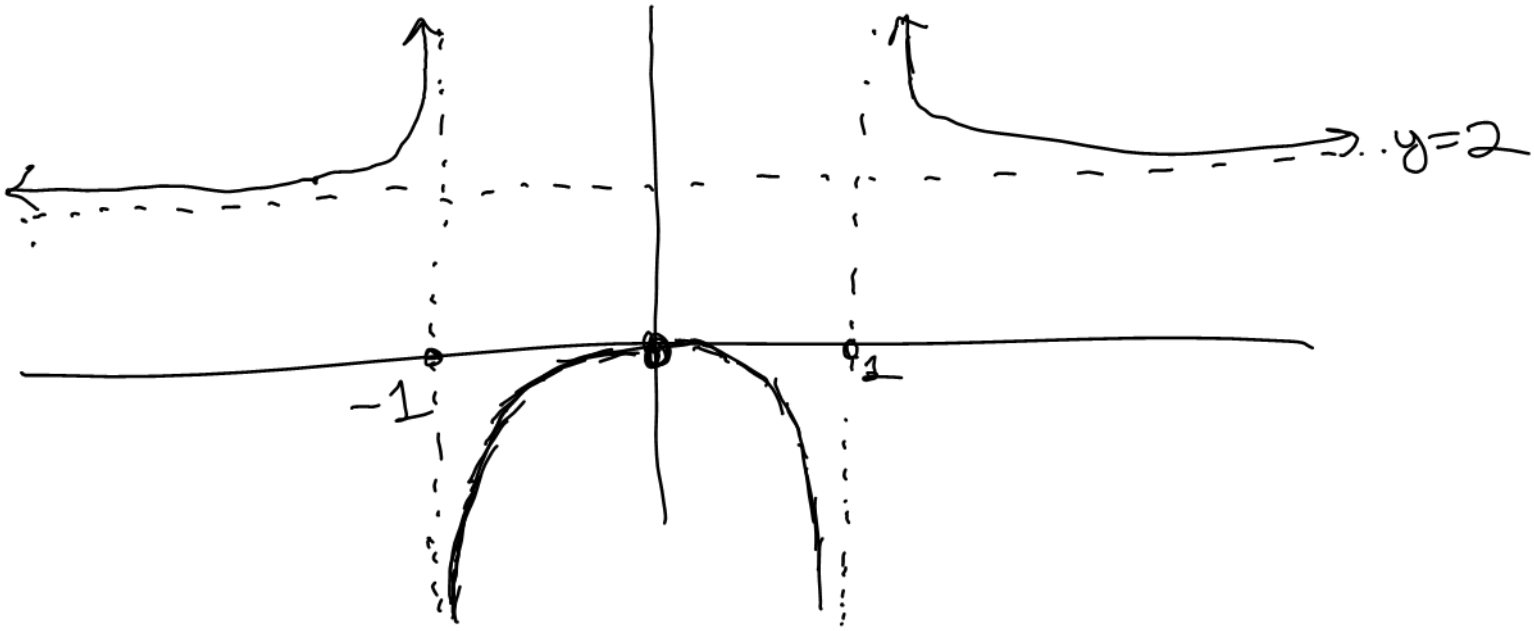
No pos. loc. for an IP. However, Concavity could change at the asymptote, so we include it.



Test Points	-100	$-\frac{1}{2}$	$\frac{1}{2}$	100
Sign of f'	+	+	−	−
Sign of $12x^2+4$	+	+	+	+
Sign of $(x^2-1)^3$	+	−	−	+
Sign of f'' ☆	+	−	−	+
I/D Test	Inc	Inc ↑	↓ Dec	Dec
C Test	CU	CD	CD	CU

① Rel. max at $(0, f(0)) = (0,0)$

② Concavity does change at the asymptotes


y=2
-1
0
1

Example 3. Sketch a graph of $f(x) = \dfrac{x^3}{x^2+1}$ $\qquad \dfrac{x^3}{x^2+1}$

Domain: $(-\infty, \infty)$

Intercepts: Only passes through the origin

Asymptotes:

VA: $x^2+1=0$ NO SOLUTION

$\Rightarrow$ No VAs

HAs: None b/c we have a slant asy.

$$x^2+1 \overline{)\, x^3} \quad \leftarrow x$$

$$-(x^3+x)$$

$$-x$$

$y = x$ is our slant asymptote

$$f'(x) = \frac{x^2(x^2+3)}{(x^2+1)^2}$$

$x^2 \geq 0$, $(x^2+3) \geq 0$, $(x^2+1)^2 \geq 0$

$\Rightarrow$ f is always increasing by the I/D Test

Example 4. Sketch a graph of $f(x) = \dfrac{10x^3}{x^2 - 1}$

Example 5 (Bell Curve)**.** Sketch a graph of $f(x) = e^{-x^2}$

Example 6. Sketch a graph of $f(x) = \ln(4 - x^2)$